

MindSpark — Revision Implementation Record

Date: 2026-06-26 **Scope:** "REVISION NEEDED" feature request (19 items) implemented across 4 phases. **Status:** Complete · TypeScript clean · production build green (34 routes) · local DB migrated & seeded.

Approaches locked with the client

Decision	Choice
Quiz content	DB-backed question bank (seeded from <code>lib/quizContent.ts</code> , which stays as offline fallback)
"Classroom" → "Section"	UI labels only — DB table <code>classrooms</code> / column <code>classroom_id</code> unchanged
Teacher/Student login	App = student, Web = teacher (runtime Capacitor detection)
Parent notifications	Fully removed
Keys (rescue mechanic)	Derived from completed levels (no new table)

Phase 1 — Foundation (cleanup, rename, CSV, question bank)

Database

- New column `users.full_name VARCHAR(120)`.
- New table `quiz_questions` (level, phase, category, prompt, options JSON, correct_index, is_active, created_by).
- New table `level_deadlines` (student_id, level_number, due_at, set_by).
- Seeded all 50 original questions; each level/phase carries 5 distinct categories; `blank` → `_____`.

Parent notifications removed (req #2)

- Deleted `app/api/teacher/students/[id]/notify-parent/route.ts` and `lib/email.ts`.
- Stripped parent email / notify code from `app/api/progress/route.ts`, `app/api/teacher/students/route.ts`, `app/api/teacher/students/bulk/route.ts`, `app/api/teacher/students/[id]/route.ts`, `app/teacher/students/page.tsx`, `app/teacher/dashboard/page.tsx`.
- `parent_email` DB column kept but unused (non-destructive).

Classroom → Section (req #3)

- Every user-visible label changed to "Section" across teacher layout, dashboard, students, sections (classrooms) page, reports, and API messages. DB identifiers untouched.

CSV = Full Name + Username (req #4)

- CSV template/columns now `Full Name, Username` (student email dropped, password auto-generated as `username + "123"`).
- Single-enroll form and student edit panel now use **Full Name** instead of email/parent.

Teacher question bank (req #19)

- CRUD API `app/api/teacher/questions/route.ts` + `app/api/teacher/questions/[id]/route.ts`.
- New page `app/teacher/questions/page.tsx` (filter by level/phase, add/edit/activate/delete, mark correct answer, category autocomplete).
- "Question Bank" added to teacher nav + dashboard quick links.

Phase 2 — Gameplay (`components/game/QuizAdventure.tsx`, `app/api/quiz/route.ts`)

- **Shuffle + backup pool + categories (reqs #6, #7, #10):** game fetches a fresh set per attempt from `/api/quiz?level=N`; `lib/questionBank.ts` `sampleQuizSet()` picks one question per category, shuffles question order and answer options, and draws from the backup pool so sets aren't memorizable.
- **Unlock once → auto-progress (req #5):** teacher unlock is the entry gate; clearing a level auto-opens the next.

- **Storyline first (req #9):** new `story` screen before each level (`lib/storyContent.ts`) — narrative + enemy + "answer 5 questions first," then a **Begin Quest** button.
 - **Keys / rescue (req #17):** key counter on the map + "earned the X Key, freed Y" beat on level complete (keys derived from completed levels).
 - **Enemy death (req #15):** "💀 {Boss} Defeated!" banner + greyed/dimmed boss sprite when HP hits 0.
 - **Avatar fade (req #16):** hero desaturates/fades per wrong answer; fully grey + "Faded out!" when all are wrong.
 - **Blank** → `___` (req #8): applied at seed, on teacher input, in the API mapper, and as a render guard.
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Phase 3 — Insight & control (`lib/proficiency.ts`)

- **Per-lesson insight (req #11):** `app/api/teacher/students/[id]/insight/route.ts` returns per-category (lesson) accuracy by joining attempts to the question bank.
 - **Score** → **proficiency (req #13):** accuracy mapped to Beginning / Developing / Proficient / Advanced.
 - **Recommendation (req #14):** plain-language recommendation from band + weakest lessons.
 - Surfaced in the student panel (`app/teacher/students/page.tsx`) and the reports table (`app/teacher/reports/page.tsx`).
 - **Per-level deadlines (req #12):** `app/api/teacher/students/[id]/deadline/route.ts` + student-panel UI (set/clear, overdue flag).
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Phase 4 — Platform

- **Login split (req #1):** `lib/appTarget.ts` `isStudentApp()` detects the Capacitor app at runtime (the app loads the live web URL per `capacitor.config`, so build-time env alone can't distinguish). `components/auth/LoginForm.tsx` blocks teachers in the student app; `components/auth/TeacherAccessLink.tsx` hides teacher sign-up there. Override: `NEXT_PUBLIC_APP_TARGET=student`.
 - **Tablet (req #18):** viewport (`device-width`, no forced zoom) and responsive `dvh`/`clamp`/breakpoint layout confirmed; verified via production build (not on-device).
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Files

Created

```
database/migrations/2026-06-revision.sql
database/seed-questions.sql
types/quiz.ts
lib/questionBank.ts
lib/storyContent.ts
lib/proficiency.ts
lib/appTarget.ts
app/api/quiz/route.ts
app/api/teacher/questions/route.ts
app/api/teacher/questions/[id]/route.ts
app/api/teacher/students/[id]/insight/route.ts
app/api/teacher/students/[id]/deadline/route.ts
app/teacher/questions/page.tsx
components/auth/TeacherAccessLink.tsx
```

Modified

```
database/schema.sql
database/schema-hostinger.sql
.env.example
app/api/progress/route.ts
app/api/teacher/students/route.ts
app/api/teacher/students/[id]/route.ts
app/api/teacher/students/bulk/route.ts
app/api/teacher/classrooms/route.ts
app/login/page.tsx
app/teacher/layout.tsx
app/teacher/dashboard/page.tsx
app/teacher/students/page.tsx
app/teacher/classrooms/page.tsx
app/teacher/reports/page.tsx
components/game/QuizAdventure.tsx
components/auth/LoginForm.tsx
```

Deleted

```
app/api/teacher/students/[id]/notify-parent/route.ts  
lib/email.ts
```

⚠ Deployment — run on remote DBs (Railway / Hostinger)

Local XAMPP is migrated & seeded. Remote databases are **not** yet. Run on each:

```
mysql -u <user> -p <dbname> < database/migrations/2026-06-revision.sql  
mysql -u <user> -p <dbname> < database/seed-questions.sql
```

(Or import both via phpMyAdmin.) Without this the question bank, deadlines, and full-name features will error in production.

For a dedicated student web build, set `NEXT_PUBLIC_APP_TARGET=student`. The native app is detected automatically.

How to test locally

Requires XAMPP MySQL running. `npm run dev`, then:

- **Teacher (web):** Question Bank (add/edit), student panel (Insights + Level Deadlines), Reports (proficiency + recommendation).
- **Student:** story intro → shuffled 5-question quiz → boss with defeat banner → avatar fade on wrong → key/rescue reward → next level auto-opens.

Known limitations

- Tablet support verified by responsive design + build, **not** on a physical device/emulator.
- Per-lesson insight only counts attempts made **after** this change (older attempts used non-numeric question IDs).
- Changes are uncommitted — in the working tree for review.